

Transition: DS + algorithms

Sets recap  
Induction



First:

Set proofs!

# Induction

A very powerful proof method, used for statements of the form:

$$\forall n, P(n)$$

Strategy:

① Show  $P(1)$  is true

② Show  $\forall k > 1, P(k-1) \rightarrow P(k)$

Why?

## Alternatives:

Show true for all numbers directly.

Great if it works!

But → why is  $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ ?

Sometimes, hard to know how to tackle

→ this is just another method!

# Writing inductive proofs

3 pieces:  $\forall n, P(n)$

- Base case

- Inductive Hypothesis

- Inductive step

Recap:  $\forall n, \sum_{i=1}^n i = \frac{n(n+1)}{2}$

Side bar: why should we care?

Proof: by induction on  $n$

Base case:

Ind Hyp:

Ind Step:

Geometric series:

$$\text{if } r \neq 0, \quad \sum_{i=0}^n a \cdot r^i = \frac{a \cdot r^{n+1} - a}{r-1}$$

Proof: Fix  $r \neq 0$ , & induct on  $n$ :

Claim:  $n^3 - n$  is divisible by 3 whenever  $n$  is a positive integer.

Proof: Induction on  $n$

Next time:

Induction not on numbers

(I delayed the next 2 readings  
so we can do more induction.)